

Acid-Base Equilibria

Brønsted–Lowry Acid and Base

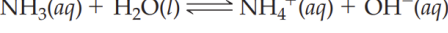
- A Brønsted–Lowry **acid** must have at least one removable (acidic) proton (H⁺) to donate.
- A Brønsted–Lowry **base** must have at least one nonbonding pair of electrons to accept a proton (H⁺)

What Is Different about Water?

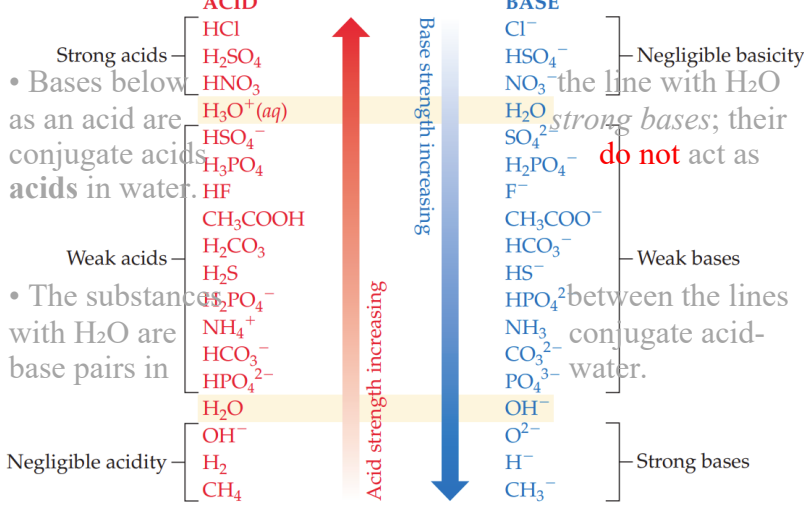
- Water can act as a Brønsted–Lowry base and accept a proton (H⁺) from an acid.
- It can also donate a proton and act as an acid.
- This makes water **amphiprotic**.

Conjugate Acids and Bases

- The term **conjugate** means “joined together as a pair.”
- Reactions between acids and bases always yield their conjugate bases and acids



with H2O as a base are **strong acids**; their conjugate bases **do not** act as **bases** in water.



Acid and Base Strength

- In every acid-base reaction, equilibrium favors transfer of the proton from the stronger acid to the stronger base to form the weaker acid and the weaker base.

Autoionization of Water

- Water is **amphoteric**. In pure water, a few molecules act as bases and a few act as acids, which is **autoionization**.

Ion Product Constant

- The equilibrium expression is $K_c = [H_3O^+][OH^-]$
 - This special equilibrium constant is referred to as the **ion product constant** for water, K_w . At 25 °C, $K_w = 1.0 \times 10^{-14}$.
- Aqueous Solutions Can Be **Acidic**, **Basic**, or **Neutral**.

pH is a method of reporting hydrogen ion concentration.

$pH = -\log[H^+]$, $pH + pOH = pK_w = 14.00$ **Measure pH**

- **Indicators**, including litmus paper, are used for less accurate measurements; an indicator is one color in its acid form and another color in its basic form.

- **pH meters** are used for accurate measurement of pH; electrodes indicate small changes in voltage to detect pH.

Strong Acids

- HCl, HBr, HI, HNO3, H2SO4, HClO3, HClO4.
- These are, by definition, strong electrolytes and exist totally as ions in aqueous solution. $HA + H_2O \rightarrow H_3O^+ + A^-$
- So, for the monoprotic strong acids, $[H_3O^+] = [acid]$.

Weak Acids

- Hydrofluoric acid (HF), Acetic acid (CH3COOH), Formic acid (HCOOH), Benzoic acid (C6H5COOH), Carbonic acid (H2CO3), Phosphoric acid, (H3PO4).
- For a weak acid, the equation for its dissociation is $HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$
- There is an equilibrium constant related to it, called the acid-dissociation constant, $K_a = [H_3O^+][A^-] / [HA]$
- The greater the value of K_a , the stronger is the acid.

Comparing Strong and Weak Acids

- Strong acids **completely** dissociate to ions. $[H^+]_{eq} = [HA]_{init}$
- Weak acids **only partially** dissociate to ions. $[H^+]_{eq} < [HA]_{init}$
- This creates a difference in conductivity and in **rates of chemical reactions**.

Polyprotic Acids

- Polyprotic acids have more than one acidic proton.
- It is easier to remove the 1st proton than any successive H⁺.
- If the factor in the K_a values for the 1st and 2nd dissociation has a difference of 3 or greater, the pH generally depends *only* on the 1st dissociation.

Factors that Affect Acid Strength

- H-A bond must be polarized with δ^+ on the H atom and δ^- on the A atom
- Bond strength: Weaker bonds can be broken more easily, making the acid stronger.
- Stability of A⁻: More stable anion means stronger acid.

Strong Bases

- The soluble hydroxides, which are the alkali metal and heavier alkaline earth metal hydroxides (Ca²⁺, Sr²⁺, Ba²⁺).
- These substances dissociate completely in aqueous solution. $MOH(aq) \rightarrow M^+(aq) + OH^-(aq)$
- The equilibrium constant is the base dissociation constant. $K_b = [M^+][OH^-] / [MOH]$

Two Main Types of Weak Bases

- 1) Neutral substances with an atom that has a nonbonding pair of electrons that can accept H⁺ (like ammonia and the **amines**)
- 2) Anions of weak acids

Relationship between K_a and K_b $K_a \times K_b = K_w$

Anions

- Anions of strong acids are neutral. e.g., Cl⁻ will not react with water.
- Anions of weak acids are conjugate bases, so they create OH⁻ in water; e.g., $C_2H_3O_2^- + H_2O \rightleftharpoons HC_2H_3O_2 + OH^-$
- Protonated anions from polyprotic acids can be acids or bases: If $K_a > K_b$, the anion will be acidic; if $K_b > K_a$, the anion will be basic.

Cations

- Group I/II (Ca²⁺, Sr²⁺, Ba²⁺) metal cations are neutral.
- Polyatomic cations are typically the conjugate acids of a weak base; e.g., NH4⁺.
- Transition and post-transition metal cations are acidic for there are no H atoms in these cations.

Hydrated Cations

- Transition and post-transition metals form hydrated cations. The water attached to the metal is more acidic than free water molecules, making the hydrated ions acidic.

Oxyacids (e.g., Cl-OH, Br-OH, H-OH)

- Consist of H2O and one other element, which is a nonmetal
- Generally, the larger electronegativity of the nonmetal, the acidity increases for acids with the same structure.

Oxyacids with Same “Other” Element

- If an element can form more than one oxyacid, the oxyacid with more O atoms is more acidic; e.g., sulfuric acid versus sulfurous (HSO3) acid.

If the oxidation number increases, the acidity increases.



Carboxylic Acids

- CAs are organic acids containing the -COOH group.
- Factors contributing to their acidic behavior:
 - Other O attached to C draws electron density from O-H bond, increasing polarity.
 - Its conjugate base (carboxylate anion) has resonance forms to stabilize the anion.

Binary Acids

- Binary acids consist of H and one other element.
- Within a group, **H-A bond strength** is generally the most important factor.

- Within a period, **bond polarity** is the most important factor to determine acid strength.

Ionic Equilibria

Buffered solutions (缓冲溶液) are

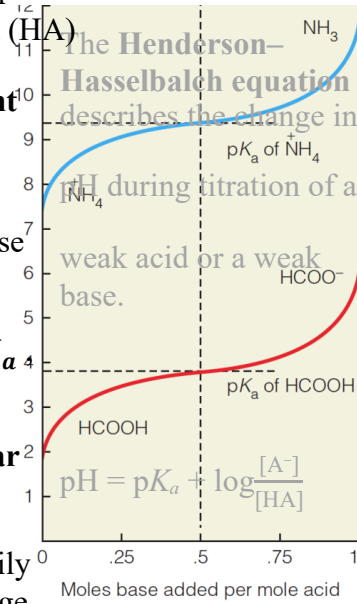
able to minimize the change in pH following addition of acid or base because the conjugate acid (HA) and conjugate base (A⁻) of the buffering compound **are present in sufficient** concentration to combine with the added H⁺ or OH⁻ and neutralize them.

Buffer solutions function because the pH of a weak acid–base solution is least sensitive to added acid or base **near the pK_a** where the conjugate acid and conjugate base of the buffer are both present in nearly **equimolar concentrations**.

- Organisms must maintain the pH inside cells and in most bodily fluids within the narrow pH range of about 6.5 to 8.0.

- The dihydrogen phosphate–hydrogen phosphate system, with a pK_a of 6.86, plays a major role in controlling **intracellular** pH because phosphate is abundant in cells.
- **Blood** contains dissolved CO2 as a waste product of metabolism, therefore the carbonic acid–bicarbonate system provides considerable buffering capacity.

CH4 Neither acid nor base	NH3 Weak base K _b = 1.8 × 10 ⁻⁵	H2O	HF Weak acid K _a = 6.8 × 10 ⁻⁴
SiH4 Neither acid nor base	PH3 Very weak base K _b = 4 × 10 ⁻²⁸	H2S Weak acid K _a = 9.5 × 10 ⁻⁸	HCl Strong acid
		H2Se Weak acid K _a = 1.3 × 10 ⁻⁴	HBr Strong acid



Basic Concepts of Chemical Bonding

Chemical Bonds

- Three basic types of bonds
 - **Ionic**: Electrostatic attraction between ions.
 - **Covalent**: Sharing of electrons.
 - **Metallic**: Metal atoms bonded to several other atoms.

The octet rule: When forming compounds, atoms tend to gain, lose, or share electrons until they're surrounded by 8 valence electrons.

- Atoms tend to lose (metals) or gain (nonmetals) electrons to make them isoelectronic to the noble gases.

Energetics of Ionic Bonding

- Making ions (ionization energy and electron affinity).
- It takes energy to convert the elements to atoms, create a cation. (endothermic)
- Energy is released by making the anion, forming solid (exothermic).

Lattice Energy

- That huge, exothermic transition is the reverse of the lattice energy – the energy required to completely separate a mole of a solid ionic compound into its gaseous ions.

- The energy associated with electrostatic interactions is governed by Coulomb’s law: $E_{el} = \kappa \frac{Q_1 Q_2}{d}$

Factors influence Lattice energy

- increasing charge on the ions
- decreasing size of ions

Covalent Bonding

- Electrostatic interactions in these bonds: attractions between electrons and nuclei; repulsions between electrons; repulsions between nuclei.
- For a bond to form, the attractions must be greater than the repulsions.

Electronegativity

- DEF: the ability of an atom in a molecule to attract electrons to itself.
- On the periodic table, electronegativity generally increases as you go
 - from left to right across a period.
 - from the bottom to the top of a group.

Electronegativity and Polar Covalent Bonds

- When two atoms share electrons unequally, a polar covalent bond results.
- Electrons tend to spend more time around the more electronegative atom. The result is a partial negative charge (not a complete transfer of charge). It is represented by δ^- . The other atom is “more positive,” or δ^+ .

Dipoles

- When two **equal, opposite** charges are separated by a distance, a dipole forms.
- A dipole moment μ , is calculated: $\mu = Qr$. It is measured in debyes (D).

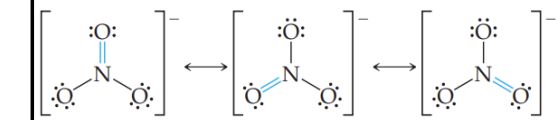
Formal charge is the charge an atom would have if all of the electrons in a covalent bond were shared equally. Formal charge = valence electrons – ½ (bonding electrons) – all nonbonding electrons

The dominant Lewis structure

- is the one in which atoms have formal charges closest to zero.
- puts a negative formal charge on the most electronegative atom.

Resonance

Localized electrons are specifically on one atom or shared between two atoms: **Delocalized electrons** are shared by multiple atoms.

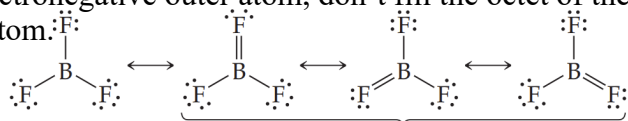


Percent ionization = $[H_3O^+]_{eq} / [HA]_{init}$

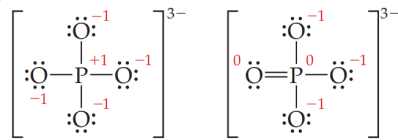
Basic Concepts of Chemical Bonding

Exceptions to the Octet Rule $\text{N}=\ddot{\text{O}}$ and $\text{N}=\ddot{\text{O}}$

- ions or molecules with an **odd number of electrons**,
- unstable and reactive
- ions or molecules with **less than an octet**,
- Elements in the second period before C can make stable compounds with fewer than 8 electrons.
- If filling the octet of the central atom results in a negative charge on the central atom and a positive charge on the more electronegative outer atom, don't fill the octet of the central atom.



- ions or molecules with **more than 8 valence electrons** (an expanded octet).
- When an element is in period 3 or below in the periodic table, it can use *d*-orbitals to make more than 4 bonds.



Covalent Bond Strength is measured by determining how much energy is required to break the bond. This is called the **bond enthalpy**.

Average Bond Enthalpies

- Average bond enthalpies are positive, because bond breaking is an endothermic process.
- Note that these are averages over many different compounds; not every bond in nature for a pair of atoms has exactly the same bond energy.

Use Bond Enthalpies to Estimate Enthalpy of Reaction

- One way to estimate ΔH for a reaction is to use the bond enthalpies of bonds broken and the new bonds formed.
- Energy is added to break bonds and released when making bonds, $\Delta H_{\text{rxn}} =$

$$\sum \text{(bond enthalpies of all bonds broken)} - \sum \text{(bond enthalpies of all bonds formed)}$$

Bond Enthalpy and Bond Length

- Measure an average bond length for different bond types.
- As the **number of bonds** between two atoms increases, the bond length decreases

Molecular Geometries

Molecular Shapes

- Lewis Structures show bonding and lone pairs, but **do not** denote shape. However, it can be used to determine shapes.

What Determines the Shape of a Molecule?

- Simply put, electron pairs, whether they be bonding or nonbonding, repel each other.
- By assuming the electron pairs are placed as far as possible from each other, predict the shape of the molecule.
- This is the **Valence-Shell Electron-Pair Repulsion** model.
- The best arrangement of a given number of electron domains is the one that **min the repulsions** among them.

Electron Domains

- The electrons-pointing directions is electron domains.
- 1 single/double/triple bond/lone pair = 1 electron domain

Molecular Geometries

- Once you have determined the electron-domain geometry, use the arrangement of the **bonded atoms** to determine the **molecular geometry**.

Bond Angle

- Nonbonding pairs are physically larger than bonding pairs
- Double and triple bonds have larger electron domains than single bonds.

- Therefore, their **repulsions are greater**. This tends to compress bond angles and have larger bond angles.

Trigonal Bipyramidal Electron Domain

- 2 distinct positions in this geometry: – Axial – Equatorial
- Lone pairs occupy equatorial positions. (an equatorial domain experiences less repulsion than an axial domain)

Shapes of Larger Molecules

Look at the geometry about **each atom** rather than the molecule as a whole.

Number of electron domains	4	3	4
Electron-domain geometry	Tetrahedral	Trigonal planar	Tetrahedral
Predicted bond angles	109.5°	120°	109.5°

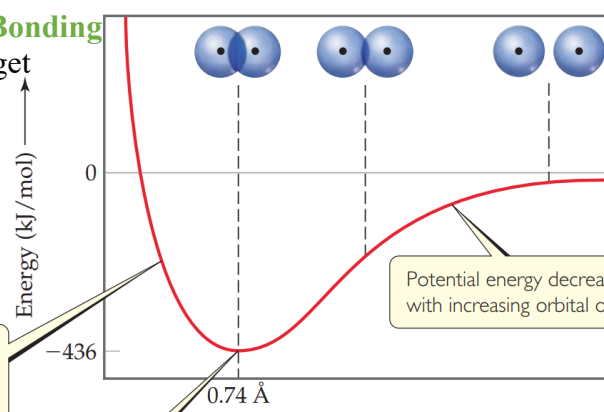
Valence-Bond Theory

- Electrons of two atoms begin to occupy the same space.
- This is called “overlap” of orbitals.
- The sharing of space between two electrons of **opposite spin** results in a covalent bond.

Overlap and Bonding

- Atoms can't get too close because the internuclear repulsions get too great.

Nucleus–nucleus repulsion increasingly important



sp Orbitals

- Mixing the s and p orbitals yields two degenerate orbitals that are hybrids of the two orbitals.
- These sp hybrid orbitals have two lobes like a p orbital.
- One of the lobes is larger and more rounded, as is the s orbital.

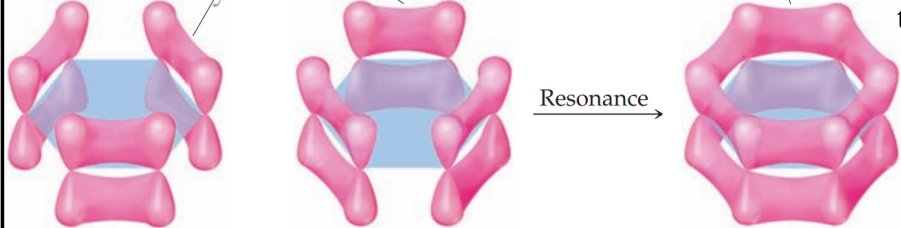
Hypervalent Molecules

- The elements which have more than an octet
- Valence-Bond model would use d orbitals to make more than 4 bonds.
- This view works for period 3 and below.
- Theoretical studies suggest that the energy needed would be too great.

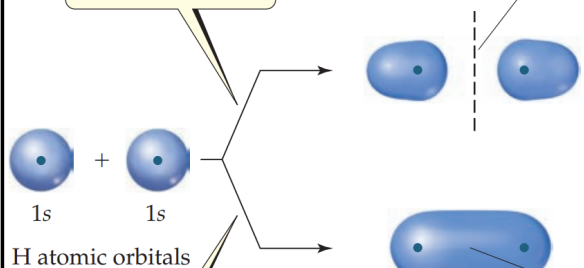
Hybridized orbitals can't form a double or triple bond, use the orbitals which are not hybridized, getting a “side-ways” overlap.

Sigma (σ) and Pi (π) Bonds

- σ bonds are characterized by head-to-head overlap.
- cylindrical symmetry of electron density about the internuclear axis.
- π bonds are characterized by side-to-side overlap.
- electron density above and below the internuclear axis.



Destructive combination leads to antibonding H_2 molecular orbital

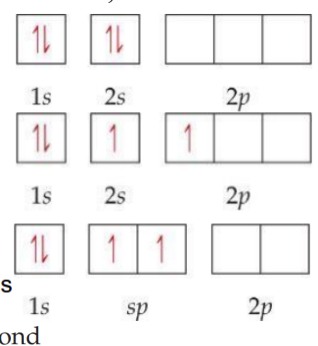


Constructive combination leads to bonding H_2 molecular orbital

Hybrid Orbitals

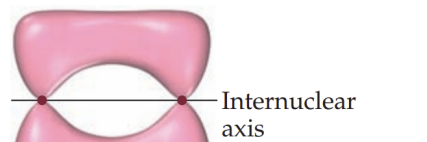
- To **explain molecular geometries**, we **assume** that the atomic orbitals on an atom mix to form new orbitals called **hybrid orbitals**.
- **Hybrid orbitals** form by “mixing” of atomic orbitals to create new orbitals of equal energy, called degenerate orbitals.
- When two orbitals “mix” they create two orbitals; when three orbitals mix, they create three orbitals; etc.

the orbital diagram for a ground-state Be atom



envision to “promote” one 2s e- to a 2p orbital

hybridize one s orbital and one p orbital to form two degenerate sp hybrid orbitals



Bonding in Molecules

- Single bonds are always σ -bonds.
- Double bonds have one σ -bond, one π -bond.
- Triple bonds have one σ -bond, two π -bonds.

Benzene

The organic molecule benzene (C_6H_6) has six σ -bonds and a p orbital on each C atom, which form delocalized bonds using one electron from each p orbital.

Molecular Orbital (MO) Theory

- Wave properties are used to describe the energy of the electrons in a molecule.
- **Molecular orbitals** have many characteristics like atomic orbitals:
 - maximum of two electrons per orbital
 - Electrons in the same orbital have opposite spin.
 - Definite energy of orbital
 - Can visualize electron density by a contour diagram

More on MO Theory

- They differ from atomic orbitals because they represent the entire molecule, not a single atom.
- Whenever two atomic orbitals overlap,

2 molecular orbitals are formed: one bonding, one antibonding.

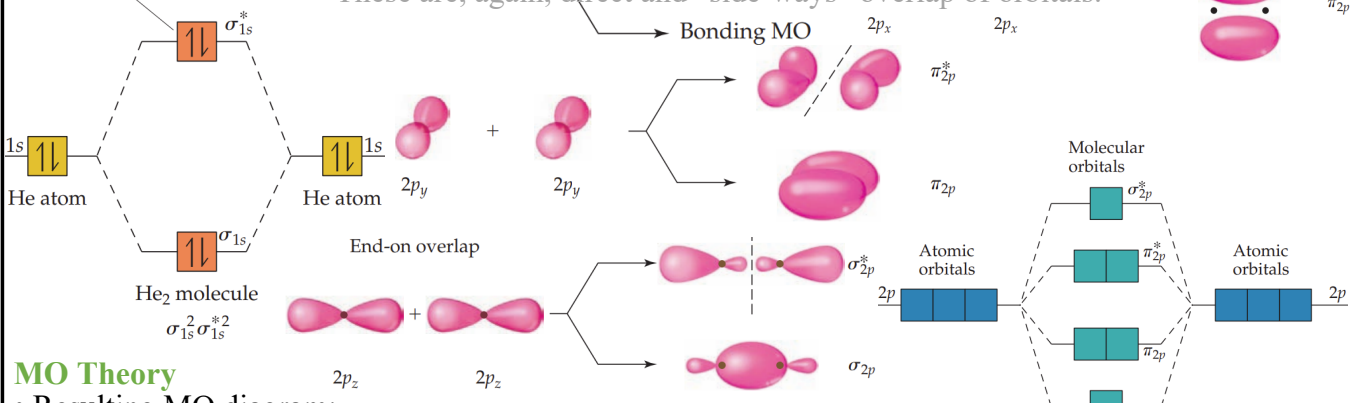
- Bonding orbitals are constructive combinations of atomic orbitals.
- Antibonding orbitals are destructive combinations of atomic orbitals. They have a new feature unseen before: A nodal plane occurs where electron density equals zero.

MO Diagram

- An energy-level diagram, or MO diagram shows how orbitals from atoms combine to give the molecule.
- In H_2 the 2 electrons go into the bonding molecular orbital (lower in energy).
- Bond order = $\frac{1}{2}(\text{\# of bonding electrons} - \text{\# of antibonding electrons})$

s and p Orbitals Can Interact

- For atoms with both s and p orbitals, there are two types of interactions:
 - The s and the p orbitals that face each other overlap in σ fashion.
 - The other 2 sets of p orbitals overlap in π fashion.
- These are, again, direct and “side-ways” overlap of orbitals.



MO Theory

- Resulting MO diagram:
 - There are σ and σ^* orbitals from s and p atomic orbitals.
 - There are π and π^* orbitals from p atomic orbitals.
 - Since direct overlap is stronger, the effect of raising and lowering energy is greater for σ and σ^* .

Molecular Geometries-Continue s and p Orbital Interactions

Increasing 2s-2p interaction

- In some cases, s orbitals interact with the p_z orbitals more than p_x and p_y orbitals.
- It raises the energy of the p_z orbital and lowers the energy of the s orbital.

σ_{2s} - σ_{2p} interaction causes energy splitting. σ_{2p} goes to higher energy, σ_{2s} goes to lower energy.

• The p_x and p_y orbitals are degenerate orbitals.

Molecular orbital energy

O_2, F_2, Ne_2 B_2, C_2, N_2

MO Diagrams for Diatomic Molecules of 2nd Period Elements

Large 2s-2p interaction			Small 2s-2p interaction		
B ₂	C ₂	N ₂	O ₂	F ₂	Ne ₂
σ_{2p}^*			σ_{2p}^*		
π_{2p}^*			π_{2p}^*		
σ_{2p}			π_{2p}		
π_{2p}			σ_{2p}		
σ_{2s}^*			σ_{2s}^*		
σ_{2s}			σ_{2s}		

MO Diagrams and Magnetism

- **Diamagnetism** is the result of all electrons in every orbital being spin paired. These substances are weakly repelled by a magnetic field.
- **Paramagnetism** is the result of the presence of one or more unpaired electrons in an orbital. **Paramagnetism of Oxygen**
- Lewis structures would not predict that O_2 is paramagnetic.
- The MO diagram clearly shows that O_2 is paramagnetic.
- Both show a double bond (bond order = 2).

Heteronuclear Diatomic Molecules

- Diatomic molecules can consist of atoms from different elements.
- Atomic orbitals have different energy, so the interactions change slightly.
- The more electronegative atom has orbitals lower in energy, so the bonding orbitals will more resemble them in energy.

Biomacromolecules of Life – Nucleic Acid

- **Nucleotides:** building blocks of biopolymers, nucleic acids.

α -D-Ribofuranose β -D-Ribofuranose

D-Ribose

Note: The absolute configurations remain the same, although the designation of R/S configurations may change.

α -D-Ribopyranose β -D-Ribopyranose

Ribose and deoxyribose structures

see β -D-Ribofuranose

Two types of **sugar pucker** most commonly found in nucleic acids. **C3'-endo:** Prevalent in A-form DNA & RNA duplexes. **C2'-endo:** Characteristic of B-form DNA duplexes.

Base structures: The 2 types of heterocyclic bases are derivatives of **purine** and **pyrimidine**.

Pyrimidine

Uracil (2,4-Dioxypyrimidine) Thymine (2,4-Dioxo-5-methylpyrimidine) Cytosine (2-Oxo-4-aminopyrimidine)

Purine

Adenine (6-Aminopurine) Guanine (2-Amino-6-oxopurine)

Cytidine Deoxycytidine

Hydrogen bond donors and acceptors

An exocyclic amine group **can't** be a hydrogen bond acceptor.

Base Ribonucleoside Deoxy- Ribonucleotide

Adenine (A) Adenosine (AMP);adenylate^a

Guanine (G) Guanosine Guanosine

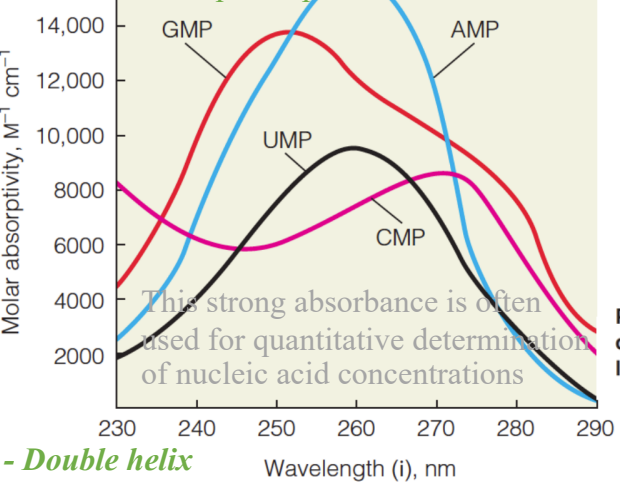
Cytosine (C) Cytidine 5'-monophosphate(GMP);guanylate^a

Uracil (U) Uridine (CMP);cytidylate^a

Thymine (T) Deoxythymidine or thymidine (UMP);uridylate^a

Nucleoside structure: A nitrogenous base with a ribose (核糖) **ribonucleoside** (核糖核苷) or deoxyribose (脱氧核糖) **deoxyribonucleoside** (脱氧核糖核苷) and a phosphate (磷酸盐)

Ultraviolet absorption spectra of ribonucleotides



- Double helix

RNA contains the sugar ribose, DNA has the deoxyribose

Directionality: 5' and 3'; **Primary structure:** The sequence

Nucleic acids can be linear or circular

- Base pairing produces a regular structure in which one strand is complementary to the other. **A-T, G-C**
- Base-pair interactions lead to the formation of a double helix with stacked base pairs.

	A Form	B Form
Direction of helix rotation	Right	Right
Number of residues per turn (n)	11	10
Rotation per residue ($= 360^\circ/n$)	33°	36°
Rise ^a in helix per residue (h)	0.255 nm	0.34 nm
Pitch ^a of helix ($= nh$)	2.8 nm	3.4 nm

- Nucleosome and chromatin

Histone octamer: Two copies of H2A, H2B, H3, and H4

Nucleosome core particle: Histone octamer and 146 bp of DNA

Nucleosome: A nucleosome core particle plus histone H1 and linker DNA (about 54 bp). **Histone H1** binds to the core particle and to linker DNA

- The histone octamer is shaped like a flat disk (with a width of about 5 nm), with positively charged grooves to accommodate the **sugar-phosphate** backbone of DNA (about 146 bp) wraps around the histone octamer.
- Formation of nucleosome core particle reduces the **DNA end-to-end distance**.
- Specific **lysine (K) residues** can be acetylated. Acetylation decreases the net positive charge of the histone
- Acetylation results in loosening up of chromatin structure

- RNA structures

• Self-complementarity in a base sequence allows a chain to fold back on itself and form an RNA secondary structure containing **stems** (with Watson-Crick G-C and A-U pairs), and **hairpins** and **junctions**.

甲基 methyl 乙基 ethyl 丙基 propyl 丁基 butyl 戊基 pentyl 己基 hexyl 庚基 heptyl 辛基 octyl 壬基 nonyl 癸基 decyl

Linear - CO_2

Trigonal planar - BF_3 Bent NO_2^-

Tetrahedral - CH_4 Trigonal pyramidal - NH_3 Bent - H_2O

Trigonal bipyramidal - PCl_5 Seesaw - SF_4 T-shaped - ClF_3 Linear - XeF

Octahedral - SF_6 Square pyramidal - BrF_5 Square planar - XeF_4

Alkene -ene

Alkyne -yne

Alcohol -ol

Ether ether

Alkyl halide or haloalkane -ide

Amine -amine

Aldehyde -al

Ketone -one

Carboxylic acid -oic acid

Ester -oate

Amide -amide

Liquids and Intermolecular Forces States of Matter

The fundamental difference between states of matter is the strength of the intermolecular forces of attraction.

Stronger forces bring molecules closer together.

Solids and liquids are referred to as the condensed phases.

State of Matter relies on the balance between the kinetic energies of the particles.

interparticle energies of attraction

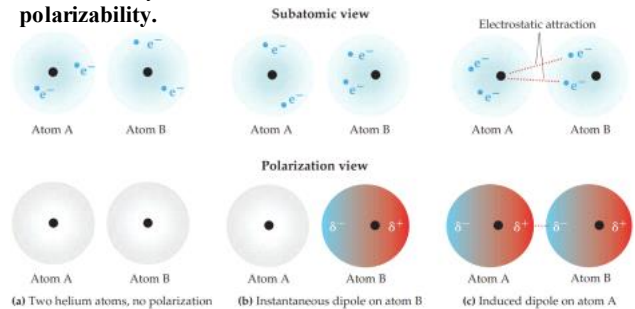
The attractions between molecules are not nearly as strong as the intramolecular attractions (bonds) that hold compounds together.

Many physical properties reflect intermolecular forces, like boiling points, melting points, viscosity, surface tension, and capillary action.

London Dispersion / van der Waals Forces

The figure below shows how a nonpolar particle (in this case a helium atom) can be temporarily polarized to allow dispersion force to form.

The tendency of an electron cloud to distort is called its polarizability.



Factors Which Affect Amount of Dispersion Force in a Molecule

- number of electrons in an atom (more electrons, more dispersion force)
- size of atom or molecule/molecular weight
- shape of molecules with similar masses (more compact, less dispersion force)

Linear molecule—larger surface area enhances intermolecular contact and increases dispersion force

Spherical molecule—smaller surface area diminishes intermolecular contact and decreases dispersion force

n-Pentane (C₅H₁₂) bp = 309.4 K

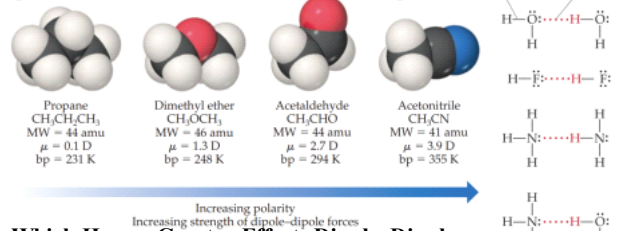
Neopentane (C₅H₁₂) bp = 282.7 K

Polarizability & Boiling Point

- Something more difficult to polarize has a lower boiling point.
- This means less intermolecular force (smaller molecule: lower molecular weight, fewer electrons).

Dipole-Dipole Interactions

- Polar molecules have a more positive and a more negative end - a dipole (two poles, δ⁺ and δ⁻).
- The oppositely charged ends attract each other.
- For molecules of approximately equal mass and size, the more polar the molecule, the higher its boiling point.



Which Have a Greater Effect: Dipole-Dipole Interactions or Dispersion Forces?

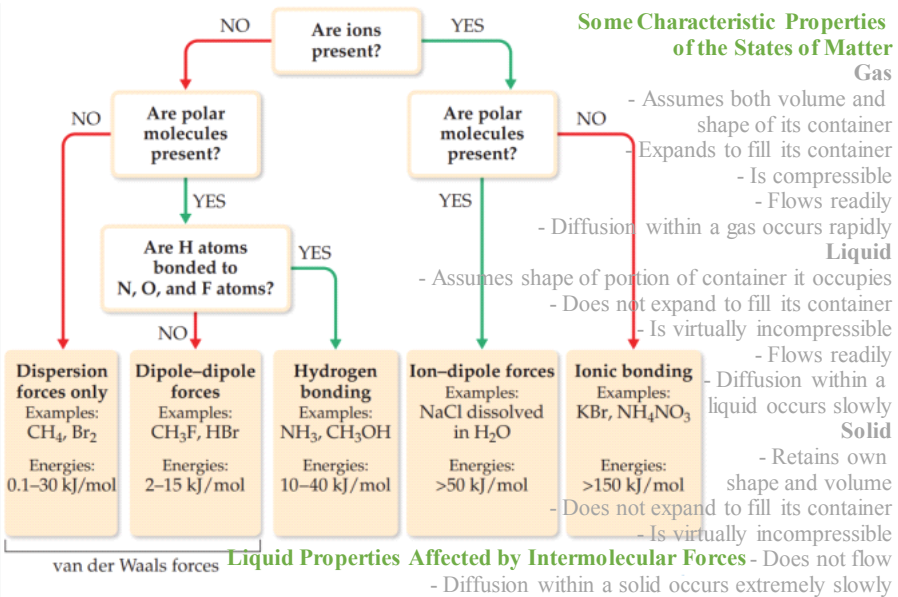
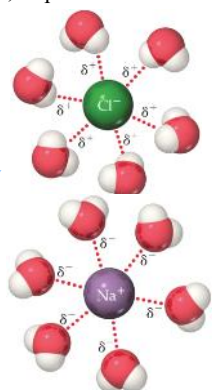
- If two molecules are of comparable size and shape, dipole-dipole interactions will likely be the dominating force.
- If one molecule is much larger than another, dispersion forces will likely determine its physical properties.

Hydrogen Bonding (special Dipole-Dipole)

- A hydrogen bond is an attraction between a hydrogen atom attached to a highly electronegative atom (N, O, and F) and a nearby small electronegative atom (a nearly bare nucleus which contains one proton) in another molecule or chemical group.

Ion-Dipole Interactions

- Ion-dipole interactions are found in solutions of ions.
- The strength of these forces is what makes it possible for ionic substances to dissolve in polar solvents.



Viscosity

- Resistance of a liquid to flow is called viscosity.
- It is related to the ease with which molecules can move past each other.
- Viscosity increases with stronger intermolecular forces and decreases with higher temperature.
- Water acts as if it has a “skin” on it due to extra inward forces on its surface. Those forces are called the surface tension.

Cohesion and Adhesion

- Intermolecular forces that bind similar molecules to one another are called cohesive forces, which attract the liquid to itself.
- Intermolecular forces that bind a substance to a surface are called adhesive forces, which attract the liquid to the wall of the tube.
- These forces are important in capillary action (The rise of liquids up narrow tubes).

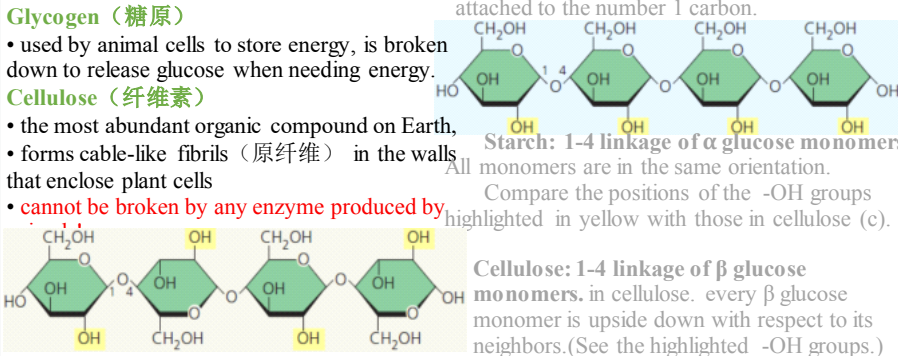
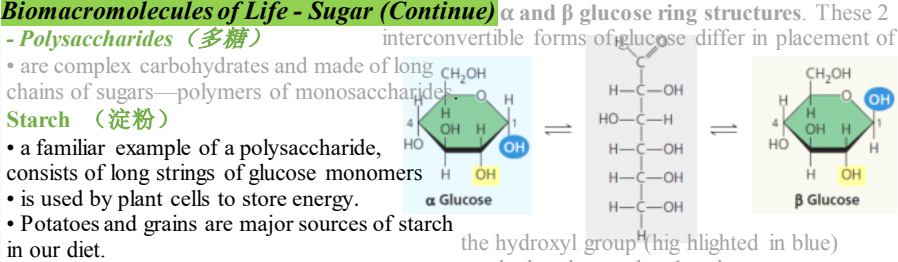
Biomacromolecules of Life - Protein

Amino acids

- In the structural formula, the symbol R means one side chain.
- 20 standard amino acids, all have primary amino group and one carboxylic acid bonded to the same carbon, C_α (Exception: proline).
- At neutral pH, amino acids exist as an internal salt.
- Natural carbohydrate: D-isomer
- Natural amino acid α-carbon is usually a chiral center (Exception: glycine).
- All natural amino acids are defined as L-amino acids (Exception: glycine).
- Almost all natural amino acid α-carbons have the S configuration (Exception: glycine and cysteine)

Classification of Amino Acids

- Side chains of amino acids are responsible for many of the unique properties of proteins.
- 3 major classes of side chains: - Nonpolar (hydrophobic) - Polar - Charged
- The 20 different amino acids are commonly abbreviated with 1- or 3-letter codes.



Is a Compound Ionic or Covalent?

- Simplest approach: Metal + nonmetal-ionic; nonmetal + nonmetal-covalent.
- Exceptions: higher oxidation numbers (Electrons assigned to atom): can give covalent bonding.
- Electronegativity difference: still doesn't take into account oxidation number.
- Properties of compounds are best: melting points.

Valence electrons: 6 4 6 6 4 6

(Electrons assigned to atom): 6 4 6 7 4 5

Formal charge: 0 0 0 -1 0 +1

Organic and Biological Chemistry

General Characteristics of Organic Molecules

- Carbon makes four bonds. • C—H are most common.
- C forms stable (strong) bonds with many elements, including C, H, O, N, and the halogens. • **functional groups**: groups of atoms that determine how an organic molecule.
- Most prevalent bonds are nonpolar C—C and C—H; solubility in water is low for many organic compounds.
- Organic molecules, such as glucose, that have polar groups are soluble in polar solvents.

Molecules with long nonpolar regions and polar regions act as surfactants, bringing **nonpolar** material into aqueous solution (used in detergents and soaps).

- Acid-Base Properties**
- Many OM's contain acidic/basic functional groups.
- Carboxylic acids (—COOH) are the most common acids.
- Amines (—NH₂, —NHR, or —NR₂, where R is an organic group made up of C and H atoms) are the most common bases.

Hydrocarbons

Alkane Ethane CH₃CH₃
Hydrocarbons consist of **ONLY C and H**.

- They are grouped based on

Alkene Ethylene CH₂=CH₂
the number of bonds between carbon atoms.

Alkyne Acetylene CH≡CH

Properties Common to Hydrocarbons

- Since they are nonpolar, they are insoluble in water but soluble in nonpolar solvents.
- Melting points and boiling points are determined by **dispersion forces** (low molar mass hydrocarbons - gases; moderate molar mass hydrocarbons - liquids; high molar mass hydrocarbons - solids).

Uses of Some Simple Alkanes

- Methane (CH₄): in natural gas (heating fuel)
- Propane (C₃H₈): in bottled gas (heating and cooking fuel)
- Butane (C₄H₁₀): in disposable lighters and fuel canisters for camping
- Alkanes with 5 to 12C atoms: gasoline

Condensed structural formulas don't show all C—H bonds. Compounds with the same molecular formula but different connections of atoms are called **structural isomers**.

When two or more substituents are present, list them alphabetically.

Systematic Nomenclature of Organic Compounds
What substituents? prefix base suffix
How many carbons? Cycloalkanes
What family? Cycloalkanes
Possible with at least 3 C, but sp³ requires 109.5°, not a very stable molecule, the same for 4.
Naming: add cyclo- as a prefix to alkane name.

Reactions of Alkanes

- Relatively unreactive due to the lack of polarity and presence of only C—C and C—H σ bonds, which are very **stable**.
- They do not react with acids, bases, or oxidizing agents.
- The most important reaction observed is combustion

Saturated vs. Unsaturated (饱和 不饱和)

Unsaturated are **more reactive** than saturated hydrocarbons.

Overlapping p orbitals create one π bond
Rotation around double bond requires considerable energy to break π bond
Rotation has destroyed orbital overlap
No free rotation about the double bond

Geometric Isomers
cis isomer
trans isomer
This end of molecule rotated 90°

- Since there is no free rotation around the double bond, the direction of the longest chain can differ for four or more C.
- Compounds that have all atoms connected to the same atoms but differ in 3-D arrangement are geometric isomers.
- cis** (same) or **trans** (opposite) isomers.

Alkynes
Naming: ends in -yne (instead of -ane or -ene)
cis-2-Butene
trans-2-Butene

Addition Reactions of Alkenes and Alkynes (加成反应)

- Add atoms to the double or triple bond, making it saturated or more saturated
- π bonds are broken, electrons form σ bonds to added atoms.
- Work with H₂ (hydrogenation), HX (hydrogen halides or water), or X₂ (halogenation)

Aromatic Hydrocarbons

- Aromatic hydrocarbons have 6-membered rings containing localized and delocalized electrons.
- The π ring is much more stable than a π bond. So much less reactive than alkenes and alkynes.
- They undergo substitution reactions (取代反应) rather than addition reactions: groups replace H on a ring (e.g., nitration, halogenation, alkylation).

Aromatic Nomenclature

- Many aromatic hydrocarbons are known by their common names.
- Others are named as derivatives of benzene.

The chemistry of an OM is largely determined by the **functional groups** it contains.

- Contain one or more -OH group (the alcohol group or the hydroxyl group)
- Named from the parent hydrocarbon; the suffix is changed to -ol and a number designates the C to which the -OH group is attached

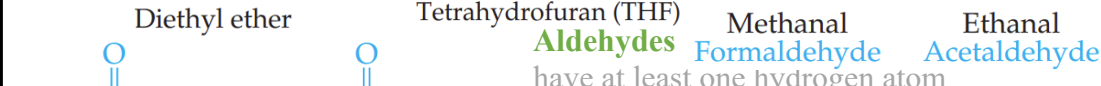
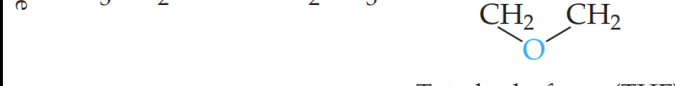
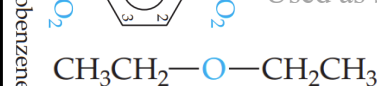
Properties and Uses of Alcohols

- Polar molecules**: lead to water solubility and higher boiling points

- Methanol**: used as a fuel additive
- Ethanol**: in alcoholic beverages
- Glycerol**: cosmetic skin softener and food moisturizer
- Ethylene glycol**: in antifreeze
- Phenol**: making plastics and dyes; topical anesthetic in throat sprays
- Cholesterol**: important biomolecule in membranes, but can precipitate and form gallstones or block blood vessels

Ethers

- R—O—R
- Formed by dehydration between alcohol molecules
- Not very reactive (except combustion)
- Used as solvents for organic reactions.



Important Aldehydes and Ketones

- Many aldehydes are **natural flavorings**: vanilla, cinnamon, spearmint, and caraway.
- Ketones are used extensively as **solvents**; the most important solvent other than water is acetone, which dissolves in water and dissolves many organic compounds.

Carboxylic Acids

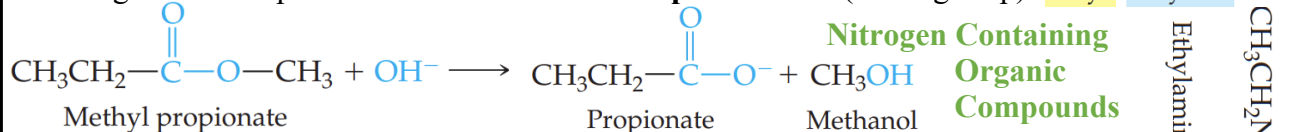
- Structure: hydroxyl group bonded to the carbonyl group
- H on the hydroxyl group is weakly acidic.
- Important in manufacturing polymers for films, fibers, and paints
- Oxidation product of alcohols (some make aldehydes)

Esters

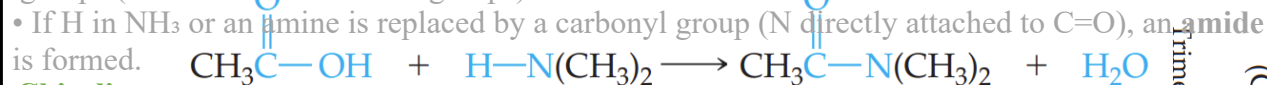
- Esters are the products of reactions between carboxylic acids and alcohols, and are found in many fruits and perfumes.
- Naming: name the alcohol part as an alkyl name; separate word is the acid part as an -ate anion.

Decomposition of Esters

- Heating an ester in presence of an **acid** catalyst and water can decompose the ester. (This is the reverse reaction of the preparation of an ester; it is an equilibrium.)
- Heating an ester in presence of a **base** results in **saponification** (making soap).



- Amines are organic derivatives of ammonia (NH₃). H atoms can be replaced by R groups (the same or different R groups).



Chirality

- Carbons with **four** different groups attached to them are **chiral**, which are optical isomers, or **enantiomers**.
- Enantiomers have the same physical and chemical properties when they react with **nonchiral** reagents.
- Enantiomers rotate plane-polarized light in opposite directions.
- Molecular recognition is carried out by some special biomolecules (e.g. receptors and enzymes) that are themselves spatially arranged in a specific direction.
- Most biomolecules possess handedness and their binding sites can thus distinguish between the **right-handed** or **left-handed** compounds.

Chiral Carbons Are Present in Most Biological Molecules

- The bonds formed by an asymmetric carbon can be arranged in 2 different mirror images (**stereoisomers**) of each other
- Stereoisomers** are either right-handed or left-handed and typically have completely different biological activities
- Asymmetric carbons are key features of amino acids, carbohydrates, and nucleic acids
- Biomolecules are **stereospecific**, because they are chiral and their binding sites may be recognized by different enantiomers.

